

14.8 Answers

① a) $f(\pm 1, 0) = \overset{\text{max}}{1}$, $f(0, \pm 1) = \overset{\text{min}}{-1}$

b) $f(\pm 2, 1) = 4$, $f(\pm 2, -1) = -4$

c) $f(1, 3, 5) = 70$, $f(-1, -3, -5) = -70$

d) $\frac{2}{\sqrt{3}}$, $-\frac{2}{\sqrt{3}}$

e) 1 , $\frac{1}{3}$

f) $f(1, \sqrt{2}, -\sqrt{2}) = 1 + 2\sqrt{2}$, $f(1, -\sqrt{2}, \sqrt{2}) = 1 - 2\sqrt{2}$

② $f(\pm \frac{1}{\sqrt{2}}, \mp \frac{1}{2\sqrt{2}}) = e^{\frac{1}{4}}$, $f(\pm \frac{1}{\sqrt{2}}, \pm \frac{1}{2\sqrt{2}}) = e^{-\frac{1}{4}}$

③ $(0, 0, \pm 1)$

④ $40 \times 40 \times 20 \text{ cm}$